

SVX Applications Catalog

SystemVerilog Extension Library — Applications ([apps/](#))

Version 2.1.2 beta

October 2026

Mark Glasser

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1. Sparse Memory

The `mem` application provides a simulation-friendly model of a large address space without requiring backing storage for every possible location. Memory is allocated lazily: pages and blocks are created on demand as data is written, so only those locations that actually receive writes consume simulator memory. A sparse representation makes it practical to model a 32-bit (or wider) address space in a verification environment without pre-allocating four gigabytes of simulator memory.

Address Decomposition

Every address is decomposed into three fields according to parameterized bit widths. The upper `PAGE_BITS` bits identify the *page*; the next `BLOCK_BITS` bits identify the *block* within the page; the remaining low-order bits identify the *byte* within the block. This three-level hierarchy mirrors the structure of a real memory subsystem and makes sparse allocation efficient even for large address spaces.

Security Model

The sparse memory supports read/write restrictions at three levels of granularity: page, block, and individual word. A restriction value of `RESTRICT_READ` prevents reads; `RESTRICT_WRITE` prevents writes; `RESTRICT_READ_WRITE` prevents both. Restrictions are consulted in order from coarsest (page) to finest (word); the first matching restriction wins. When a security violation occurs the offending operation is aborted and an error code is recorded.

Error Reporting

After every read or write operation the caller can inspect whether the operation succeeded. `last_operation_failed` returns 1 if an error occurred. `get_last_error()` returns the specific `error_t` value, and `get_last_op_string()` returns a human-readable string that names the operation, the address, and the reason for failure. This is particularly useful for debug and self-checking testbenches.

Bounded Memory

The `mem_bounded` class extends `mem` with configurable lower and upper address bounds. Any access outside those bounds is rejected with an `ERROR_ADDRESS_BOUNDS` error. An optional bounds lock prevents the bounds from being changed once set, which is useful for enforcing region ownership in a shared address space.

Implementation Notes

Internally the sparse memory is structured as a *map of pages*, where each page is a *map of blocks*, and each block is a *vector of bytes*. The maps use SVX's `map` class (a balanced tree), so lookup is $O(\log n)$ in the number of allocated pages or blocks. Only the `mem` and `mem_bounded` classes are intended for direct instantiation; `mem_page` and `mem_block` are internal implementation classes.

1.1 Parameters

All four classes share the same parameter set.

Parameter	Default	Description
ADDR_BITS	32	Total address space width in bits.
PAGE_BITS	16	Width of the page index field.
BLOCK_BITS	8	Width of the block index field within a page.
WORD_SIZE	4	Word size in bytes; words must be naturally aligned.

1.2 Types

Key typedefs (defined in `mem_types.svh` and `mem_base`):

Type	Description
<code>addr_t</code>	Parameterized address type: bit [ADDR_BITS-1:0].
<code>word_t</code>	Word type: bit [WORD_SIZE*8-1:0].
<code>byte_t</code>	8-bit unsigned byte.
<code>pri_t</code>	Priority type (<code>uint32_t</code>), used by <code>pri_queue</code> , not <code>mem</code> .
<code>restrict_t</code>	Security restriction enum (see below).
<code>error_t</code>	Error code enum (see below).
<code>operation_t</code>	Last operation enum (see below).

`restrict_t`

Value	Meaning
RESTRICT_NONE	No restriction; reads and writes are allowed.
RESTRICT_READ	Reads are blocked; writes are allowed.
RESTRICT_WRITE	Writes are blocked; reads are allowed.
RESTRICT_READ_WRITE	Both reads and writes are blocked.

`error_t`

Value	Meaning
ERROR_NONE	No error; the last operation succeeded.
ERROR_PARAMETERS_WRONG	Class parameters are inconsistent (fatal).
ERROR_MISALIGNMENT	Word access to a non-word-aligned address.
ERROR_PAGE_SECURITY_VIOLATION	A page-level restriction blocked the access.
ERROR_BLOCK_SECURITY_VIOLATION	A block-level restriction blocked the access.
ERROR_WORD_SECURITY_VIOLATION	A word-level restriction blocked the access.
ERROR_ADDRESS_BOUNDS	Address is outside the configured bounds (<code>mem_bounded</code> only).

`operation_t`

Value	Meaning
OP_NONE	No operation has been performed yet.
OP_READ	The last operation was a word read.
OP_WRITE	The last operation was a word write.
OP_READ_BYTE	The last operation was a byte read.
OP_WRITE_BYTE	The last operation was a byte write.

1.3 Class: mem

The top-level user-facing class. Declare and instantiate it directly:

```
mem#(ADDR_BITS, PAGE_BITS, BLOCK_BITS, WORD_SIZE) m = new();
```

Method	Returns	Arguments	Description
write	void	addr_t addr word_t data	Write a word to addr. Addr must be word-aligned.
read	word_t	addr_t addr	Read and return the word at addr. Returns 0 if the location was never written.
write_byte	void	addr_t addr byte_t data	Write a single byte to addr.
read_byte	byte_t	addr_t addr	Read and return the byte at addr. Returns 0 if never written.
set_page_restriction	void	addr_t addr restrict_t r	Apply restriction r to the entire page that contains addr.
get_page_restriction	restrict_t	addr_t addr	Return the restriction on the page containing addr.
clear_page_restriction	void	addr_t addr	Remove any page-level restriction for the page containing addr.
set_block_restriction	void	addr_t addr restrict_t r	Apply restriction r to the block containing addr.
clear_block_restriction	void	addr_t addr	Remove any block-level restriction for the block containing addr.
set_word_restriction	void	addr_t addr restrict_t r	Apply restriction r to the word at addr.
clear_word_restriction	void	addr_t addr	Remove any word-level restriction at addr.
is_readable	bit	addr_t addr	Return 1 if addr may be read (no active read restriction).
is_writable	bit	addr_t addr	Return 1 if addr may be written (no active write restriction).
get_addr_restriction	restrict_t	addr_t addr	Return the effective restriction at addr (coarsest level that applies).
last_operation_failed	bit	—	Return 1 if the most recent operation encountered an error.

Method	Returns	Arguments	Description
<code>get_last_error</code>	<code>error_t</code>	—	Return the <code>error_t</code> code from the most recent operation.
<code>get_last_op_string</code>	<code>string</code>	—	Return a human-readable string describing the last operation and any error.
<code>dump_security</code>	<code>void</code>	<code>addr_t addr=0</code>	Print the security map for all pages (and blocks if allocated).
<code>dump</code>	<code>void</code>	<code>addr_t addr=0</code>	Print the contents of all allocated pages and blocks.

1.4 Class: `mem_bounded`

`mem_bounded` extends `mem` with lower and upper address bounds. Any access outside the configured range is rejected with `ERROR_ADDRESS_BOUNDS`. All methods of `mem` are available; the following are added or overridden.

Method	Returns	Arguments	Description
<code>set_bounds</code>	<code>void</code>	<code>addr_t lower</code> <code>addr_t upper</code>	Set both the lower and upper bounds in one call.
<code>set_lower_bound</code>	<code>void</code>	<code>addr_t lower</code>	Set the lower bound only.
<code>set_upper_bound</code>	<code>void</code>	<code>addr_t upper</code>	Set the upper bound only.
<code>get_lower_bound</code>	<code>addr_t</code>	—	Return the current lower bound.
<code>get_upper_bound</code>	<code>addr_t</code>	—	Return the current upper bound.
<code>set_bounds_lock</code>	<code>void</code>	<code>bit lock</code>	Lock the address bounds when <code>lock=1</code> ; further calls to <code>set_bounds*</code> are silently ignored.
<code>get_bounds_lock</code>	<code>bit</code>	—	Return 1 if the bounds are currently locked.

2. Memory Map

The `mem_map` application models a hierarchical address space of the kind found in hardware block specifications: registers, bit fields, sub-regions, memories, and overlapping views, all organized into a tree whose nodes carry precise offset and size information.

Overview

Each node in the hierarchy is a *memory space*, represented by the abstract base class `mem_space`. Concrete subclasses capture the five distinct space types recognized in hardware design: `MEMORY`, `REGISTER`, `FIELD`, `REGION`, and `VIEW`. Every space has a name, a parent, an offset (relative to the parent), and a size. The hierarchy is built by constructing child spaces and passing the parent as a constructor argument; the child registers itself with its parent automatically.

Space Types and Hierarchy Rules

The five space types form a strict containment hierarchy. Not every type may be nested inside every other type:

- A **FIELD** is a bit field within a register. Fields are leaves: they may not contain children.
- A **REGISTER** models a memory-mapped register. Its only legal children are `FIELD`s.
- A **MEMORY** represents a contiguous block of memory. Memories are leaves: they may not contain children.
- A **REGION** is a container for other spaces. It may contain `REGISTER`s, `MEMORIES`, nested `REGIONS`, and `VIEWS`, but not `FIELD`s directly. Children of a region may not overlap each other.
- A **VIEW** is like a `REGION`, but sibling `VIEWS` *are* permitted to overlap each other. This models hardware designs where the same physical addresses can be interpreted in different ways depending on a mode bit.

Address Calculation and Validation

After the hierarchy is fully constructed, call `calculate_and_check()` on the root to propagate absolute addresses down the tree and to check for overlapping siblings. `get_addr()` and `get_end_addr()` then return the absolute address range of any node.

Navigation

`find_addr(addr)` searches the hierarchy for the deepest node whose range contains `addr` and returns it. `find_addr_all(addr)` returns a list of *all* nodes (at every level) whose range contains `addr`, which is useful when `VIEWS` overlap.

Implementation

`mem_space` extends SVX's `tree` class. Children are stored in an SVX `deque` and accessed through a typed iterator. The overlap check in `check()` iterates over all pairs of non-VIEW siblings and verifies that their address ranges do not intersect.

2.1 Space Types

Type	Leaf?	Allowed children
FIELD	Yes	None
REGISTER	No	FIELD only
MEMORY	Yes	None
REGION	No	REGISTER, MEMORY, REGION, VIEW
VIEW	No	REGISTER, MEMORY, REGION, VIEW (may overlap sibling VIEWS)

2.2 Class: `mem_space` (Abstract Base)

All concrete space classes extend `mem_space#(ADDR_SIZE=32)`. The methods below are defined on the base and inherited by all subtypes. Pure-virtual methods (`get_type()`, `check_child()`) are implemented by each concrete subclass.

Method	Returns	Arguments	Description
<code>get_offset</code>	<code>addr_t</code>	—	Return the offset of this space relative to its parent.
<code>get_size</code>	<code>mem_size_t</code>	—	Return the size of this space in address units.
<code>get_addr</code>	<code>addr_t</code>	—	Return the absolute base address (valid after <code>calculate_and_check()</code>).
<code>get_end_addr</code>	<code>addr_t</code>	—	Return the absolute end address (<code>base + size - 1</code>).
<code>get_type</code>	<code>mem_space_type_t</code>	—	Return the space type (FIELD, REGISTER, MEMORY, REGION, or VIEW). Pure virtual.
<code>get_type_name</code>	<code>string</code>	—	Return the space type as a human-readable string.
<code>check_child</code>	<code>bit</code>	<code>mem_space_t child</code>	Return 1 if child is a legal child of this space type. Pure virtual.
<code>insert_space</code>	<code>void</code>	<code>mem_space_t child</code>	Insert child into this space; emits a warning and skips insertion on type violation.
<code>get_error</code>	<code>bit</code>	—	Return 1 if this space has been flagged as erroneous.

Method	Returns	Arguments	Description
<code>clear_error</code>	<code>void</code>	—	Clear the stored error message.
<code>calculate_and_check</code>	<code>bit</code>	—	Propagate absolute addresses and check for overlapping siblings. Call on the root after the hierarchy is fully built.
<code>find_addr</code>	<code>mem_space_t</code>	<code>addr_t addr</code>	Find and return the deepest space whose range contains <code>addr</code> .
<code>find_addr_all</code>	<code>list_t</code>	<code>addr_t addr</code>	Return a list of all spaces (at every level) whose range contains <code>addr</code> .
<code>dump</code>	<code>void</code>	—	Print the hierarchy rooted at this node, with offsets, sizes, and absolute addresses.

2.3 Classes: `mem_region` and `mem_view`

`mem_region#(ADDR_SIZE=32)` and `mem_view#(ADDR_SIZE=32)` both extend `mem_space` and provide factory methods for building the hierarchy. `mem_view` additionally permits sibling instances to overlap in address space. Constructor: `new(name, parent, offset, size)`.

Method	Returns	Arguments	Description
<code>add_register</code>	<code>void</code>	<code>string name</code> <code>addr_t offset</code> <code>mem_size_t size</code>	Create and attach a <code>mem_register</code> child at the given offset and size.
<code>add_memory</code>	<code>void</code>	<code>string name</code> <code>addr_t offset</code> <code>mem_size_t size</code>	Create and attach a <code>mem_memory</code> child at the given offset and size.
<code>add_region</code>	<code>void</code>	<code>string name</code> <code>addr_t offset</code> <code>mem_size_t size</code>	Create and attach a nested <code>mem_region</code> child.
<code>add_view</code>	<code>void</code>	<code>string name</code> <code>addr_t offset</code> <code>mem_size_t size</code>	Create and attach a <code>mem_view</code> child.

2.4 Class: `mem_register`

`mem_register#(ADDR_SIZE=32)` extends `mem_space`. Its only legal children are `FIELDs`. Constructor: `new(name, parent, offset, size)`.

Method	Returns	Arguments	Description
<code>add_field</code>	<code>void</code>	<code>string name</code> <code>addr_t offset</code> <code>mem_size_t size</code>	Create and attach a <code>mem_field</code> child representing a bit field at the given bit offset and width.

2.5 Classes: `mem_memory` and `mem_field`

Both are leaf nodes with no children and no additional methods beyond those inherited from `mem_space`. They exist primarily as type tags so that `check_child()` can enforce the containment rules. Constructor for both: `new(name, parent, offset, size)`.

3. Priority Queue

The `pri_queue` application is a generic priority queue that supports two fundamental operations: `push()` and `pop()`.

Overview

`push(pri, item)` inserts `item` into the queue at priority level `pri`. `pop()` removes and returns the item with the highest priority. When multiple items share the same priority they are returned in the order they were inserted — first-in, first-out within each priority level.

Implementation

The underlying data structure is a *map of queues*: an SVX `map` whose keys are priority values and whose values are SVX `queue` instances holding the items at each priority. Because SVX's `map` is implemented as an ordered tree, the entry with the highest key (priority) is retrieved in constant time via `last()`. When the last item at a given priority is popped, the corresponding map entry is deleted automatically, keeping the map compact.

Type Parameters

`pri_queue` is doubly parameterized: `T` is the item type (default `int`) and `P` is the traits class (default `void_traits`). The priority type `pri_t` is a `uint32_t` defined in `pri_queue_pkg`. `pri_queue` extends SVX's `typed_container#(T,P)` and inherits its `is_empty()`, `size()`, and `clear()` methods.

3.1 Class: `pri_queue`

Declared as `pri_queue#(type T=int, type P=void_traits)` in `pri_queue_pkg`. Constructor: `new()` — no arguments required.

Method	Returns	Arguments	Description
<code>push</code>	<code>void</code>	<code>pri_t pri</code> <code>T item</code>	Insert item into the queue at priority level <code>pri</code> . Higher values of <code>pri</code> represent higher priority.
<code>pop</code>	<code>T</code>	—	Remove and return the item with the highest priority. When multiple items share the top priority, returns the one inserted earliest. Returns <code>P::empty</code> if the queue is empty.
<code>is_empty</code>	<code>bit</code>	—	Return 1 if the queue contains no items.
<code>size</code>	<code>size_t</code>	—	Return the total number of items across all priority levels.
<code>clear</code>	<code>void</code>	—	Remove all items from the queue.